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V-23—Application of limits of error for volume devices

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1.0 Purpose

The purpose of this bulletin is to clarify when acceptance limits of error (LOE) apply and when in-service LOE apply.

2.0 Scope

This bulletin applies to all volumetric devices when its performance is being evaluated for the purpose of determining compliance with the *Weights and Measures Act*.

3.0 References

In section 2 of the *Weights and Measures Regulations*, the use of acceptance LOE is based on the conditions surrounding the inspection including: the initial inspection of the device; approval evaluations; repair to the measuring elements following failure of the device to measure within the applicable LOE; and, within 30 days after the time of the last two conditions. In addition, sections in the regulations specific to LOE for devices state that the use of acceptance LOE apply to tests made under "controlled conditions".

4.0 Terminology

Acceptance and in-service LOE are the LOE as defined in section 2 of the *Weights and Measures Regulations*.

4.1 Initial Inspection refers to the inspection required by subsection 8(b) of the *Weights and Measures Act*. It is the inspection of approved measuring machines, devices and ancillary equipment prior to their first use in trade.

5.0 Background

5.1 There is not unanimous agreement that the controlled conditions as outlined in Measurement Canada's Bulletin V-3 can be easily met or measured in the field without measurement instrumentation (thermometers, pressure gauges, etc.) nor is there any current data to support the widening or narrowing of the scope of these conditions.

5.2 At the time of writing this bulletin, Measurement Canada is reviewing the current LOE tables applicable to measuring devices against other international standards with the intent of clarifying and simplifying the tables in the future. It has been noted that the Canadian acceptance LOE are possibly the tightest in the world which must be rationalized, given the extreme environment devices are used in this country. It is anticipated that the Canadian limits of errors will at some point have to be aligned with other legal metrology organizations around the globe.

6.0 Policy

Due to the uncertainty as to whether controlled conditions can be achieved in the field, acceptance and in-service LOE will be applied as follows:

6.1 Acceptance LOE apply to all initial inspections of measuring devices in a factory setting and to devices being evaluated for approval purposes, including those done in the field.

6.2 In-service LOE apply to all inspections conducted in the field other than those for approval purposes.

7.0 Revision

The purpose of revision 1 was to revise the bulletin format. Minor editorial changes were implemented to suit the format.

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